

CONTROLLED ENGINEERING ANALYSIS

KaSiHub Shares Page and Complete Journey

Frontend surfaces, API-by-API gateway map, backend orchestration, data authority, payment settlement, incorporation, certificate generation and failure paths

Control	Value
Repository revision	89ac2cbe024e45cd63cf7d85f521ac8ce785b4a3+working-tree
Branch	Klaasvaakie/crypto-certificate-safe-release
Analysis date	2026-08-31
Evidence basis	Static source/migration inspection plus a local production-build browser test of member login, controlled API failure, visible retry and multi-phase recovery. No deployed runtime or provider state was asserted.

1. Executive conclusion

There is no single shares journey. The repository contains three user-facing paths that converge only after controlled backend processing:

Surface	User	Primary route	Authority
Member portfolio	Existing ecosystem member	Dashboard SharesView	Encore shares ledger through the normal ecosystem session
Applicant/shareholder portal	Presale applicant or issued shareholder	/shares/account	Presale-scoped session plus order/application/KYC state and read-through to shares ledger
Private acquisition	Invited applicant	/presale or /presale/[invite]	Invitation, application, KYC, reservation, payment evidence and incorporation workflows

The correct end-to-end chain is:

Private invite	T O	Applicant + KYC	T O	Reservation	T O	Verified settlement	T O	Controlled incorporation
----------------	--------	-----------------	--------	-------------	--------	---------------------	--------	--------------------------

Only incorporation writes the authoritative share purchase and certificate records. A reservation, submitted transaction hash, card-return page, provider process update, confirmed presale order, or generated PDF is not independently proof that shares were issued.

Key findings and remediation status

Status	Finding / change	Impact
Fixed	Member /api/shares now preserves each certificate's linked phase, paid/bonus allocation, issue price and authoritative purchase amount.	Multi-phase holdings no longer inherit the first/default phase.
Fixed	SharesView now stores bounded errors, ends loading on 4xx/5xx, and exposes a Retry portfolio control.	An upstream failure is visible and recoverable instead of becoming a permanent spinner.
Improved	Phase totals and sold shares now come from authoritative inventory metadata. Unconnected Aureus and payout projections are hidden instead of shown as zeros.	The page distinguishes known ledger facts from unavailable data.
Medium	Legacy browser-print certificate functions remain beside authoritative PDF downloads.	Two certificate experiences exist; the approved sealed-template path should remain the trusted artifact.
Structural	Certificate numbering is strongest for presale incorporation (SOL-P<phase>-<sequence>); admin reissue and direct-wallet legacy generators remain separate.	Repository-wide numbering uniformity is not yet proven.

2. System and trust-boundary map

Layer	Responsibility	Trust rule
React/Next pages	Render applicant, shareholder and member experiences; hold transient form state.	Browser state is never settlement or issuance authority.
Next route handlers	Read HttpOnly cookies, validate basic request shape, proxy bounded errors, generate certificate PDFs.	Gateway may narrow access; Encore remains business authority.
Encore identity/auth	Validate hashed session tokens and enforce ecosystem versus presale scope.	Wrong scope is denied even if a token is otherwise valid.
Encore presale	Invitation, application, reservation, payment evidence, continuation, portal and incorporation orchestration.	Presale DB is isolated from shares DB until incorporation.
Encore payments/KYC	Canonical chain checks, receiving configuration, provider reconciliation, Didit/evidence policy.	Provider evidence and KYC decisions fail closed.
Encore shares	Share phases, purchases, certificates, ranges, integrity snapshots and register queries.	Authoritative ownership source.
PostgreSQL + private bucket	Durable state, constraints, idempotency keys, evidence metadata and document bytes.	Rows and constraints outrank UI projections.

Session separation

Cookie	Scope	Created by	Used by
kasihub_session	ecosystem	Normal login or verified shareholder conversion	Member dashboard, /api/shares, member certificate route, admin routes
kasishares_session	presale	Applicant registration or applicant login	/shares/account, /presale continuation, KYC, reservation, payment recovery, presale certificate route

Both cookies are HttpOnly, SameSite=Lax, path=/, secure in production, and normally seven-day sessions. The backend stores token hashes and checks session_scope through shared access guards.

Principal data stores

Store	Core records	Critical constraints
presale	campaigns, invitations, applications/versions/declarations, orders, payments/events, email delivery, incorporation batches	inventory bounds; active-owner uniqueness; idempotency; unique provider references; legal state checks
shares	phases, purchases, certificates, adjustments, phase sequences, distinctive-number sequence	non-negative inventory; unique operation/certificate links; non-overlapping allocation discipline; complete integrity snapshot
identity	users, profiles, scoped sessions	presale/ecosystem scope; revoked and expired sessions excluded
kyc + documents bucket	cases, Didit status, private evidence metadata and bytes	profile ownership; type/size policy; admin-only review
payments	obligations, intents, attempts, events, receiving configuration, custody evidence	state machine, dedupe and canonical evidence checks

3. Journey A - existing member SharesView

Dashboard opens	T O	GET /api/shares	T O	GET phases + my shares	T O	Adapter projects data	T O	Portfolio renders
-----------------	--------	-----------------	--------	------------------------	--------	-----------------------	--------	-------------------

Frontend behavior

Step	Component behavior	Result / caveat
1	SharesView reads <i>currentMember</i> from the client store.	A missing profile now produces a bounded session error and Retry control.
2	Calls GET /api/shares?memberId=<currentMember.id> with no-store.	The browser-supplied ID is not trusted by Encore; profile access is rechecked.
3	Displays active presale campaign cards from GET /api/presale/campaigns .	Public summary excludes receiving address/token contract and remains invitation-bound.
4	Renders hero metrics, active/retracted certificates, authoritative phase inventory, current value, purchase amount, and links to /presale.	Unavailable payout/Aureus data is not presented as an authoritative zero.
5	Certificate actions either use browser-print legacy HTML or protected PDF download routes, depending on control.	Approved sealed PDF path has stronger holder/integrity checks.

Member API chain

Browser / Next API	Encore API	Authentication	Transformation
GET /api/shares?memberId=...	GET /shares/phases	kasihub_session	Maps authoritative total/available inventory, sold count, price, status and BOGO; tolerates the previous response shape during rolling deployment.
same request	GET /shares/me/:profileId	kasihub_session + requireProfileAccess	Preserves certificate phase, paid/bonus split, purchase amount and issue price; separates active and revoked records.
GET /api/presale/campaigns	GET /presale/campaigns	Public summary	Shows name, issuer, class, inventory, price and BOGO; no payment instructions.
GET /api/shares/certificates/:number	GET /profiles/me then GET /shares/me/:profileId	kasihub_session	Selects only signed-in holder certificate; seals or derives PDF data.
GET /api/shares/certificates/verify/:verificationId	GET /shares/certificates/verify/:verificationId	Public verification	Returns bounded verified/not-found response with no-store.
POST /api/shares/buy	none	kasihub_session	Always 409: direct share purchases disabled; redirects policy to private presale.

Failure behavior

The member gateway now returns a bounded error while preserving authorization status and sends **Cache-Control: private, no-store**. SharesView renders an explicit failure card, does not estimate holdings while the register is unavailable, and retries the same authenticated request. A local production-build browser run proved: login -> KasiShares -> controlled 503 -> visible error/retry -> recovered two-phase portfolio.

4. Applicant and shareholder account journey

Open /shares/account	T O	Applicant login	T O	Portal aggregate	T O	Resume/recover	T O	Certificate/account
----------------------	--------	-----------------	--------	------------------	--------	----------------	--------	---------------------

Portal state machine

State	Portal presentation	Allowed action
No session	Separate KaSiShares login form	POST /api/presale/auth/login
No application	No application to continue	Use active private invitation; optional test URL only when configured
Editable application	Step N of 5 and saved completion	Follow server-produced encrypted resume URL
Invitation unavailable	Private access no longer available	Contact support; no bypass
Reservation in progress	Order/payment status card	Recover payment; cancel only eligible unpaid order with acknowledgement
Payment submitted/detected	Crypto recovery panel	POST payment recheck; transaction hash remains preserved
Confirmed / awaiting issuance	Paid and bonus allocation shown	Wait for controlled incorporation
Issued	Authoritative holding + certificate	Download protected PDF; optionally open ecosystem account
Incorporated but certificate missing	Reconciliation required	Support/admin must reconcile; UI does not fabricate certificate
Revoked	Revoked status	No active ownership claim from that certificate

Portal API chain

Next API	Encore API	Purpose	Key control
POST /api/presale/auth/login	POST /presale/auth/login	Authenticate applicant	Sets HttpOnly kasishares_session; bounded 401 message
POST /api/presale/auth/logout	POST /presale/auth/logout	Revoke backend session and clear cookie	Cookie is cleared even if backend revoke fails
GET /api/presale/portal	GET /presale/applicant/portal	Aggregate applicant, KYC, order, continuation and shareholder holdings	requirePresaleSession; server-derived continuation
POST .../:ref/cancel	POST /presale/orders/:ref/cancel	Cancel unpaid reservation	Requires session and acknowledgeNoPaymentSent=true
POST .../:ref/payment-recheck	POST /presale/orders/:ref/payment-recheck	Re-run canonical settlement verification	Only signed-in applicant's order
GET /api/presale/certificates/:number	GET /presale/applicant/portal	Generate holder certificate PDF	Certificate must be inside signed-in portal holdings
POST /api/presale/ecosystem-account	POST /presale/shareholder/ecosystem-account	Convert verified issued shareholder into normal member account	Precondition: verified shareholder with issued shares; creates ecosystem session and pending subscription

5. Journey C - private acquisition application

Entry and invitation validation

Step	Frontend	API/backend	Failure boundary
0	Open /presale or /presale/[invite]	Server passes invite token to PresaleClient	Without valid invite, offer call fails closed; devPreview=1 is local static/read-only only
1	Fetch offer	GET /api/presale/offer?invite= -> GET /presale/offer	Invalid, expired, revoked, exhausted, inactive or out-of-window offer is denied
2	If already logged in, fetch portal	GET /api/presale/portal	Server state may restore application/order instead of starting over

Five application phases

Phase	Frontend data and action	Backend effect
1. Applicant profile	Identity/contact, applicant type, password and private invitation context	POST /api/presale/members -> registerPresaleMember creates/reuses presale user/profile, application, presale session and account email delivery
2. Investment profile	Quantity and investor context	Progress/draft saved against authenticated application; server retains versioned application state
3. Funds and banking	Source of funds, ownership, bank details, optional SWIFT/BIC	Client validates every control before phase advance; optional blank SWIFT remains valid
4. Identity + declarations	Didit hosted session/iframe; AML, suitability and accuracy declarations	KYC session and signed decision determine verified status; cannot advance merely from client choice
5. Terms + payment choice	Scroll authoritative 10-page terms; accept acknowledgement; choose Remitano USDT or WebPay card	POST order validates invitation, applicant ownership, KYC, declarations, terms version, quantity, inventory and idempotency

Application and KYC API inventory

Next API	Encore calls	Input/output
POST /api/presale/members	POST /presale/members	Profile/application registration; sets presale cookie; 200 reused or 201 created
POST /api/presale/progress	POST /presale/applicant/progress	phaseCompleted + bounded draft; completion percent returned
POST /api/presale/kyc-session	GET /profiles/me; POST /kyc/international/cases; POST /kyc/international/cases/:id/didit-session	Returns secure provider URL/session
GET /api/presale/kyc-status	GET /profiles/me; case/status queries	Returns backend-derived KYC status and verified flag
POST /api/presale/kyc-documents	GET /profiles/me; create case; POST private document bytes	identity_document PDF/JPEG/PNG or selfie JPEG/PNG; 1 byte-10 MB; duplicate-safe
POST /presale/applications	Encore direct contract	Create application draft; owner/campaign uniqueness
PATCH /presale/applications/:id/phases/:phase	Encore direct contract	Versioned phase save with optimistic concurrency and events

6. Reservation and payment - API by API

Order creation

Control	Mechanism
Idempotency	Browser generates an Idempotency-Key; gateway rejects absence; backend stores a hash unique per invitation.
Identity binding	kasishares_session identifies profile; buyer email must match; external profile and application are linked.
Eligibility	Active campaign/invitation, remaining invitation/campaign inventory, accepted current terms/declarations and approved KYC.
Pricing	USDT quote comes from campaign. WebPay uses invitation override, bounded campaign test price, or configured default in that order.
Atomic reservation	Campaign reserved_shares and invitation used_shares are incremented with order creation; deadline is persisted.
Output	Order reference, payment rail, quote, deadline and restricted payment instructions; access/resume credential returned where applicable.

Payment routes

Next API	Encore API	Rail	Authority / outcome
POST /api/presale/orders/:ref/payment-proof	POST /presale/orders/:ref/payment-proof	Remitano/USDT	Saves hash, loads canonical chain evidence, validates network/token/receiver/amount/confirmations and optional custody; status may remain pending/manual/underpaid/rejected/confirmed
POST /api/presale/orders/:ref/payment-recheck	POST /presale/orders/:ref/payment-recheck	Remitano/USDT	Re-evaluates saved hash; safe recovery without creating a second reservation
POST /presale/webhooks/usdt	receivePresalePaymentEvent	USDT event	HMAC authenticated, deduplicated event; does not trust arbitrary browser proof
POST /api/presale/orders/:ref/webpay-checkout	POST /presale/orders/:ref/webpay-checkout	WebPay card	Returns hosted form URL/fields/checksum; card data never enters KaSiHub form
POST /api/presale/webpay/notify	POST /presale/webhooks/webpay	WebPay settlement	Raw form parsed; checksum, merchant/order/amount and uniqueness checked; confirmed settlement applied
POST /api/presale/webpay/process	POST /presale/webhooks/webpay-process	WebPay process	Signed process-stage status; FAILED/REJECTED/EXPIRED/REVERSED do not allocate shares
GET /api/presale/orders/:ref	GET /presale/orders/:ref	Both	Uses X-Presale-Access-Token bearer, never query string; refreshes order status

Payment state interpretation

State	Meaning	Shares issued?
awaiting_payment	Reservation exists; no accepted payment evidence	No
payment_submitted	Hash/evidence recorded	No
payment_detected	Transfer found but confirmation/reconciliation incomplete	No
confirmed	Canonical/provider settlement requirements satisfied	Not yet - eligible for incorporation
incorporated	Presale order linked to authoritative shares transaction	Yes, if certificate/purchase transaction committed
expired/cancelled	Inventory released under controlled rules	No

7. Incorporation into the authoritative register

Confirmed orders	T O	Prepare batch	T O	Hash manifest	T O	Apply batch	T O	Reconcile result
------------------	--------	---------------	--------	---------------	--------	-------------	--------	------------------

Admin and internal API chain

API	Action	Control
GET /api/admin/presale/orders -> GET /admin/presale/orders	Review campaign orders and incorporation status	Normal admin ecosystem session + requireAdminAccess
POST /api/admin/presale/incorporation-batches -> POST /admin/presale/incorporation-batches	Select confirmed, unincorporated orders; create deterministic manifest and hash	Admin only; prepared batch evidence persisted
POST /api/admin/presale/incorporation-batches/:id/apply	Apply each order to shares domain	Admin only; batch and order idempotency; partial interruption is recoverable
POST /internal/presale/reconcile-incorporation	Retry confirmed pending orders	Internal-only scheduled recovery
POST /internal/presale/expire-orders	Expire unpaid deadlines and release inventory	Internal-only transactional maintenance
POST /internal/presale/retry-crypto-payments	Retry saved pending crypto evidence	Internal-only; preserves evidence and classification

Shares transaction

Write	Authoritative fields / invariant
share_purchases	profile, phase, paid quantity, amount, operation id, certificate id, presale order reference; unique operation and exact linkage
share_certificates	SOL-P<phase>-<sequence>, total shares, paid/bonus split, distinctive range, presale order reference, status and issue time
share_lot_sequence	Allocates one contiguous, non-overlapping distinctive range; upper design bound 1,200,000
share_certificate_phase_sequences	Transactional phase-local certificate sequence
integrity snapshot	verification id, holder/profile/address snapshots, issue price, canonical payload and SHA-256; all-or-none DB constraint
presale order	incorporation_status becomes incorporated only after authoritative write; retry counts already-incorporated orders

Failure model

If an order is marked incorporated but no matching certificate can be read, the portal emits **issuance_error** and calls for reconciliation. It does not invent a certificate. Batch retry distinguishes already-incorporated work, allowing recovery after a process dies between the shares and presale commits.

8. Certificate lifecycle and verification

Stage	Backend / route	Integrity behavior
Issue	Presale incorporation writes share purchase + certificate	Sealed holder/allocation/price/range payload and SHA-256 stored with unique verification ID
Read in member view	GET /shares/me/:profileId	requireProfileAccess; returns holder's ledger records
Download as member	GET /api/shares/certificates/:number	Gets /profiles/me then holder certificates; refuses missing/other certificate
Download as applicant	GET /api/presale/certificates/:number	Certificate must exist in signed-in portal holdings
Generate	generateShareCertificatePdf	Validates allocation sum, range length, issue price pair, integrity pair and issue date; uses approved Solidus template
Verify	GET /shares/certificates/verify/:verificationId	Recalculates payload hash and compares ledger snapshot fields; public bounded verification route
Revoke	POST /admin/shares/certificates/:number/revoke	Admin-only status transition
Reissue	POST /admin/shares/certificates/reissue	Admin-only legacy path; numbering differs from SOL presale protocol

PDF generation safeguards

Validation	Rejected condition
Identity	Blank certificate number or holder name
Quantity	Non-positive total; paid/bonus mismatch
Distinctive range	Only one endpoint present, reversed/non-positive range, or range length != total shares
Issue price	Only price/currency present or negative/non-finite value
Integrity	Only verification/hash present or hash not 64 lowercase hex
Sealed ledger	Missing snapshot member, SHA mismatch, payload/ledger mismatch, holder mismatch or issue-price mismatch
HTTP response	Private no-store attachment, sanitized filename and nosniff

Legacy certificate distinction

SharesView still contains client-side HTML print functions for KasiShare and Aureus certificates. These use current member/display values and are not the same as the approved template, sealed ledger snapshot and verification route. They should be treated as presentation legacy, not equivalent legal/ledger evidence.

9. Consolidated browser-to-backend API register

This register lists the complete API surface directly traversed by the three shares journeys. Admin and internal routes are listed separately in section 7.

Method	Next route	Encore target	Access	Purpose
GET	/api/shares	/shares/phases + /shares/me/:profileId	ecosystem	Member portfolio projection
POST	/api/shares/buy	none	ecosystem	409 direct purchase disabled
GET	/api/shares/certificates/:number	/profiles/me + /shares/me/:profileId	ecosystem	Protected PDF
GET	/api/shares/certificates/verify/:id	/shares/certificates/verify/:id	public	Integrity verification
GET	/api/presale/campaigns	/presale/campaigns	public	Safe campaign summary
GET	/api/presale/offer	/presale/offer	invite	Private offer
POST	/api/presale/members	/presale/members	invite	Applicant/profile creation
POST	/api/presale/auth/login	/presale/auth/login	credentials	Applicant login
POST	/api/presale/auth/logout	/presale/auth/logout	presale	Session termination
GET	/api/presale/portal	/presale/applicant/portal	presale	Aggregate portal
POST	/api/presale/progress	/presale/applicant/progress	presale	Draft progress
POST	/api/presale/kyc-session	profiles + KYC case + Dikit session	presale	Start/resume KYC
GET	/api/presale/kyc-status	profiles + KYC status	presale	Poll verification
POST	/api/presale/kyc-documents	profiles + KYC case/documents	presale	Private evidence upload
POST	/api/presale/orders	/presale/orders	presale + idempotency	Create reservation
GET	/api/presale/orders/:ref	/presale/orders/:ref	access token	Refresh order
POST	/api/presale/orders/:ref/payment-proof	same Encore path	presale	Submit crypto hash
POST	/api/presale/orders/:ref/payment-recheck	same Encore path	presale	Retry crypto verification
POST	/api/presale/orders/:ref/webpay-checkout	same Encore path	access token	Hosted card checkout
POST	/api/presale/orders/:ref/cancel	same Encore path	presale	Cancel eligible unpaid order
POST	/api/presale/webpay/notify	/presale/webhooks/webpay	provider checksum	Card settlement webhook
POST	/api/presale/webpay/process	/presale/webhooks/webpay-process	provider checksum	Hosted-process status
GET	/api/presale/certificates/:number	/presale/applicant/portal	presale	Applicant PDF
POST	/api/presale/ecosystem-account	/presale/shareholder/ecosystem-account	presale	Open normal member account

10. State, failure and recovery matrix

Failure	User-visible behavior	Durable evidence	Recovery
Invalid invite	Private invitation required/unavailable	Invitation/campaign state	Admin issues or restores valid invite; no client bypass
Account email delayed	Reservation/profile remains saved; warning shown	presale_email_deliveries failed/pending	Internal retry email job
KYC pending/provider unavailable	Phase 4 blocks advance; retry/check controls	KYC case, provider session/events	Resume Ddidit or controlled review
Malformed optional SWIFT	Native validation returns user to field	No invalid reservation written	Correct or clear optional value
Duplicate submit	Same reservation returned/controlled conflict	idempotency key hash	Reuse existing order
Crypto verifier unavailable	Hash preserved; pending message	payment/order evidence and reason	Manual recheck + scheduled retry
Underpaid/wrong token/receiver/network	Manual/rejected status	canonical evidence and classification	Controlled support/reconciliation; no issue
WebPay return before signed notify	Confirming payment message	process state/order	Wait for signed settlement notification
WebPay failed/reversed	Explicit failed status; no allocation	provider process status/event	New controlled payment attempt/reservation policy
Incorporation interruption	May remain awaiting/reconciliation	batch manifest, order incorporation status, shares operation id	Idempotent apply/reconcile
Certificate integrity mismatch	Generation/verification fails	sealed payload/hash + ledger fields	Admin investigation; never regenerate from untrusted browser state
Member shares upstream 4xx/5xx	Bounded portfolio-unavailable message and Retry	Bounded gateway response plus server logs; no invented holdings	Retry the authenticated request; browser recovery verified

11. Corrective action ledger

Status	Action	Acceptance evidence
Done	Repair SharesView load state: capture response body/status, render bounded error and Retry; handle missing currentMember explicitly.	Route/UI contracts plus local production browser 503/retry/recovery proof
Done	Replace /api/shares default-phase projection with certificate/order-linked phase, paid/bonus split, purchase amount and issue price from authoritative response.	Multi-phase fixture proves phase 1 + phase 2 certificate values
Open	Remove or clearly demote legacy client-side certificate print buttons; route trusted downloads through sealed PDF endpoints.	No UI implies browser HTML equals authoritative certificate
Open	Unify certificate numbering/reissue protocol across presale, admin and any wallet-purchase paths.	One documented numbering authority and migration/revocation strategy
Partial	Expose sold/reserved/outstanding metrics from authoritative aggregates instead of hardcoded zero/default projections.	Sold/outstanding are now derived from phase inventory; reservation aggregation remains separate
Open	Add an end-to-end no-money test spanning invite -> applicant -> KYC-approved fixture -> reservation -> signed provider fixture -> incorporation -> both portals -> certificate verification.	One auditable scenario with order, purchase, certificate and verification IDs

Release gate

Do not declare the shares journey operational from a frontend build, a Vercel Ready badge, an Encore check, a payment success screen, or a generated certificate alone. Prove the exact deployed revision, gateway and Encore routes, database migrations, authenticated browser journey, signed settlement callback, incorporation write, holder-specific certificate download and public integrity verification as separate gates.

12. Source index

Area	Primary sources
Member UI	src/components/views/shares-view.tsx; src/components/views/active-presale-campaigns.tsx; src/lib/shares-portfolio.ts
Applicant UI	src/app/shares/account/shares-account-client.tsx; src/app/shares/account/page.tsx
Presale UI	src/app/presale/presale-client.tsx; src/app/presale/page.tsx; src/app/presale/[invite]/page.tsx
Next gateways	src/app/api/shares/**; src/app/api/presale/**; src/app/api/admin/shares/**; src/app/api/admin/presale/**
Backend	encore/domains/shares/api.ts; encore/domains/presale/api.ts; applicant-continuation.ts; settlement.ts; shareholder-portfolio.ts; payments/**; kyc/**
Certificates	src/lib/share-certificate-pdf.ts; src/lib/share-certificate-integrity.ts; public/certificate-templates/solidus-shareholder-certificate.pdf
Tests	src/lib/shares-portfolio.test.ts; src/app/api/shares/route.test.ts; src/components/views/shares-view.contract.test.ts; encore/domains/shares/portfolio-contract.test.ts
Schema	encore/migrations/shares/1-9; encore/migrations/presale/1-15; identity scoped-session migration; payments and KYC migrations

Document status: source-mapped, regenerated and visually verified against the working-tree fix based on the revision printed above. The local production browser path is verified. Deployment, live provider credentials, live database contents and real-money behavior remain unverified.